EDS223 Homework 3

Prioritizing potential aquaculture

Eden Gonzalez

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```
# Load libraries
library(here)
library(sf)
library(terra)
library(tidyverse)
library(tmap)
library(spData)
library(kableExtra)
```

Load the data

```
# Exclusive Economic Zones (EEZs) in the West Coast, vector data
eez <- st_read(here( "data", "wc_regions_clean.shp"))
```

```
Reading layer `wc_regions_clean' from data source
  `/Users/edengonzalez/Desktop/MEDS/git/eds223-assignments/potential-aquaculture/data/wc_regi
  using driver `ESRI Shapefile'
Simple feature collection with 5 features and 5 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
Geodetic CRS:  WGS 84
```

```r
# Depth of the ocean, raster data
depth <- rast(here("data", "depth.tif"))

# All the sea surface temperature (SST) rasters from 2008-2012
sst_files <- list.files(path = "data/sst",
                        pattern = "\\.tif$",
                        full.names = TRUE)
# Create a raster stack that contains data for all five years
sst <- rast(sst_files)

# Make the depth raster have the same CRS as sst raster stack
crs(depth) <- crs(sst)
```

```r
# Verify that all CRSs match
if((st_crs(eez) == st_crs(sst)) & (st_crs(eez) == st_crs(depth) & (st_crs(sst) == st_crs(dep
  print("coordinate reference systems match!")
}else{
  warning("coordinate reference systems do NOT match")
  print(paste("eez has ", st_crs(eez)$epsg, "EPSG"))
  print(paste("depth has",  st_crs(depth)$epsg, "EPSG"))
  print(paste("sst has",  st_crs(sst)$epsg, "EPSG"))
}
```

```
[1] "coordinate reference systems match!"
```

Process data

```r
#  Mean SST from 2008-2012 in Celsius
sst_C <- mean(sst, na.rm = TRUE) - 273.15
# Crop depth raster to match the extent of the SST raster
depth_cropped <- crop(depth, sst_C)

# Resample the depth data to match the resolution of the SST data using the bilinear method a
depth_resampled <- resample(depth_cropped, sst_C, method = "bilinear")

# Check if features match
cat("After resampling:\n")
```

```
After resampling:
```

```
cat("CRS match:    ", same.crs(sst_C, depth_resampled), "\n")
```

```
CRS match:     TRUE
```

```
cat("Resolutions match:", all(res(sst_C) == res(depth_resampled)), "\n")
```

```
Resolutions match: TRUE
```

```
cat("Origins match:    ", all(round(origin(sst_C), 8) == round(origin(depth_resampled), 8)),
```

```
Origins match:     TRUE
```

```
cat("Extents match:    ", ext(sst_C) == ext(depth_resampled), "\n")
```

```
Extents match:     TRUE
```

Find suitable locations

Oysters Research has shown that oysters need the following conditions for optimal growth:

sea surface temperature: 11-30°C depth: 0-70 meters below sea level

species of interest: Kumamoto Oyster

```
# if(st_crs(suitable_sst) == st_crs(suitable_depth)){
#   print("coordinate reference systems match!")
# }else{
#   warning("coordinate reference systems do NOT match")
#   print(paste("suitable_sst has ", st_crs(suitable_sst), "EPSG"))
#   print(paste("depth has",  st_crs(suitable_sst), "EPSG"))

# }
```

Find suitable locations

Create function

```r
suitable_locations_aquaculture <- function(species, sst_min, sst_max, depth_min, depth_max =

# SST reclassification matrix
sst_rcl <- matrix(c(
  -Inf,   sst_min,  0,
  sst_min,  sst_max,  1,
  sst_max,  Inf,  0
), ncol = 3, byrow = TRUE)

suitable_sst <- classify(sst_C, rcl = sst_rcl)

# Depth reclassification matrix
depth_rcl <- matrix(c(
  -Inf,   depth_min,  0,
  depth_min,  depth_max,  1,
  depth_max,  Inf,  0
), ncol = 3, byrow = TRUE)

suitable_depth <- classify(depth_resampled, rcl = depth_rcl)

# Do not move on unless coordinate reference systems match
if(st_crs(suitable_sst) != st_crs(suitable_depth)){
  stop("coordinate reference systems do NOT match,\ntry crs(suitable_depth) <- crs(suitable_s
}


# Combine the temperature and depth rasters into one
suitable_combined <- c(suitable_sst, suitable_depth)
# Find cells that satisfy both SST and depth conditions
suitable <- lapp(suitable_combined, fun = function(temp, depth) temp * depth)


# Rasterize eez boundaries to match suitable's extent, resolution, and CRS
eez_raster <- rasterize(eez, suitable, field = "rgn_id")

# Create a raster that has cell values from suitable within the eez_raster
suitable_eez <- mask(suitable, eez_raster)
# Remove zeros
suitable_eez[suitable_eez == 0] <- NA
```

```r
# Calculate suitable area in each cell of the raster in km^2
cell_area <- cellSize(suitable_eez, unit = "km")

# Each cells in this raster contains their respective amount of suitable area in km^2
suitable_area <- cell_area * suitable_eez

# Create a raster that has the same extent, resolution, and CRS as suitable_area
regions <- rasterize(eez, suitable_area, field = "rgn_id")

# Sum of suitable area per region
suitable_eez <- zonal(suitable_area, regions, fun = "sum", na.rm = TRUE)

# Transform into a data frame
suitable_area_eez <- suitable_eez|>
  as.tibble()


# Join the suitable area per region data frame with the eez vector data
area_eez <- left_join(suitable_area_eez, eez, by = "rgn_id")


# Display table of suitable area within each eez in order by area
suitable_area_eez_table <- area_eez|>
  select(rgn_id, rgn, area) |>
  kable(col.names = c("Region Id", "Region", "Area (km^2)"), caption = paste0("Suitable Area
  kable_styling(bootstrap_options = "striped")



# Tranfrom the joined dataset into a simple feature for mapping
area_eez_sf <- st_as_sf(area_eez)


# Map the total suitable area within each eez
map <- tm_basemap("OpenStreetMap") +
tm_shape(area_eez_sf) +
  tm_polygons(fill = "area",
  fill.scale = tm_scale(
      values = c("lightsteelblue", "paleturquoise2", "paleturquoise3",
                 "cadetblue3", "seagreen4")),  # Reassign colors
    fill.legend = tm_legend(title = " Area (km^2)",
    position = tm_pos_out("right"),
```

```
              frame = FALSE,
              title.size = 0.7)) +
tm_shape(suitable_area) +
  tm_raster(col.scale = tm_scale_categorical(values = "red"),
            col.legend = tm_legend_hide()) +
  tm_add_legend(type = "polygons", fill = "red", labels = "Suitable area") +
  tm_scalebar(position = c("left", "bottom")) +  # Add scale bar
  tm_compass(position = c("right", "top")) +
  tm_credits("Source: NOAA, GEBCO, Marineregions", position = c("left", "bottom")) +
  tm_title(paste0("Suitable Area for ", species, "\nby Exclusive Economic Zone")) +
  tm_layout(frame.lwd = 0.4)

print(map)
suitable_area_eez_table

}
```

Oysters: Research has shown that oysters need the following conditions for optimal growth:

- sea surface temperature: 11-30°C
- depth: 0-70 meters below sea level

Species of interest: Pink Abolone

- sea surface temperature: 18-25°C
- depth: 0-54 meters below sea level

Species of interest: Steelhead Trout

- sea surface temperature: 7-114°C
- depth: 0-6 meters below sea level

```
suitable_locations_aquaculture("Oysters", 11, 30, -70, 0)
```

Suitable Area for Oysters
by Exclusive Economic Zone

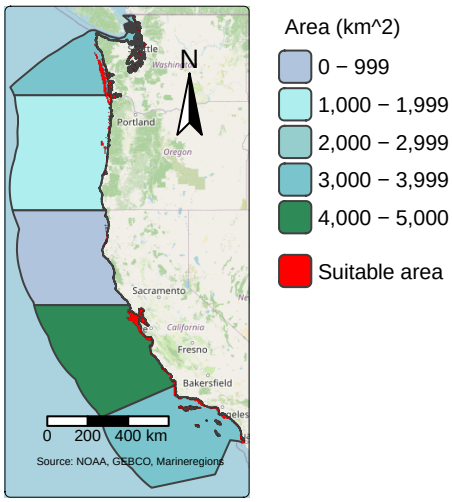


Table 1: Suitable Area per Exclusive Economic Zone, Oysters

Region Id	Region	Area (km^2)
1	Oregon	1395.7328
2	Northern California	323.9124
3	Central California	4373.5565
4	Southern California	3329.9002
5	Washington	3091.5158

```
suitable_locations_aquaculture("Pink Abolone", 18, 25, -54, 0)
```

Suitable Area for Pink Abolone
by Exclusive Economic Zone

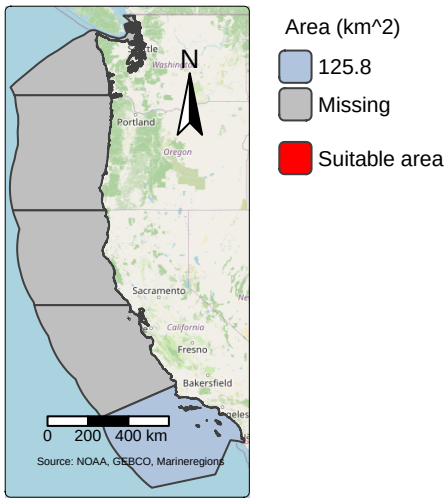


Table 2: Suitable Area per Exclusive Economic Zone, Pink Abolone

Region Id	Region	Area (km^2)
1	Oregon	NaN
2	Northern California	NaN
3	Central California	NaN
4	Southern California	125.8486
5	Washington	NaN

```
suitable_locations_aquaculture("Steelhead Trout", 7, 14, -6, 0)
```

Suitable Area for Steelhead Trout
by Exclusive Economic Zone

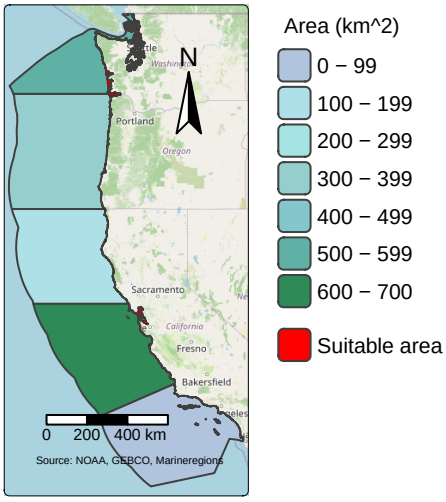


Table 3: Suitable Area per Exclusive Economic Zone, Steelhead Trout

Region Id	Region	Area (km^2)
1	Oregon	305.01881
2	Northern California	129.68478
3	Central California	629.64992
4	Southern California	35.37338
5	Washington	555.55418